14.5.1 How Good is A* Search?

If we were in a class and this was a live presentation, now your follow-up question should be: can we craft an example where A* is performing consistently worse than Dijkstra?

Turns out we can, easily: if we take the example of figure 14.20, and change just the weight of the edge from v_1 to v_G, setting it to any value smaller than 2, and at the same time we keep the same estimates, then we would be sure that A* would visit every other vertex before getting to the goal, while Dijkstra would never visit v_6 and – depending on the order it processes edges out of v_S, it might also skip vertices v_2 to v_4.

The key for this example is that the heuristic over-estimates the distance to goal for v_1 (and a few more vertices): A*, turns out, minimizes the estimated cost, which can be different from the actual cost, and whenever this estimate is pessimistic, we get in trouble.

